Higher National Diploma in Engineering(CEE)
Fourth Year, First Semester Examination - 2021
EE3211 - Electrical Machines Control & Protection

Instructions for Candidate:

Answer any **Five (05)** questions
All questions carry equal marks

No. of Questions: 06
No. of Pages: 04
Time : 3 (hours)

1.

i.) Explain the requirement of a starting method to start an induction motor. Illustrate this requirement using necessary graphs

[3 marks]

ii.) Draw the circuit diagram of auto-transformer starting method use to start an induction motor. Briefly explain the operation of this starting method

[5 marks]

iii.) State two advantages and two disadvantages of solid state based soft starter, when compared with the other starting methods use to start an induction motor

[4 marks]

iv.) A series motor needs to be used for a fan application and initially the motor shaft is coupled with a laod which provide a constant load torque. Under that condition, motor is supplied with 100V d.c supply and motor draw 5A armature current and the speed of the motor is found to be 600 r.p.m. For the fan applications, it is essential to have two speeds, 800 r.p.m and 1500 r.p.m. Speed control of the motor is achieved by varying the supply voltage to the motor using some power electronic converter (For fan applications, torque $\propto$ speed^2). Voltage required, when the speed of the fan is 600 r.p.m is found to be 150 V. Armature and field resiatnces are 0.3Ω and 0.5Ω respectively.

a.) Find the required supply voltage to obtain 800 r.p.m speed

[3 marks]

b.) Find the required supply voltage to obtain 1500 r.p.m speed

[2 marks]

c.) Calculate the armature currents for each speed

[3 marks]

[Total 20 Marks]

2.

i.) State two different dynamic breaking schemes that can be employed for three phase induction motor breaking and briefly explain each scheme

[6 marks]

ii.) A 400V, 20kW, 500 r.p.m., d.c. shunt motor is braked by plugging when running on full load. The efficiency of the motor is 74.6% and the armature resistance is 0.2 Ω and field resistance is 160 Ω;

a.) Determine the braking resistance necessary, if the maximum braking current is not to exceed twice the full-load current

[3 marks]

b.) Determine the maximum braking torque

[2 marks]

c.) Find braking torque when the motor is just reaching zero speed.

[3 marks]

iii.) Briefly explain why the supply voltage to supply frequency ratio must maintained at constant value when employed the frequency control method to vary the speed of three phase induction motor

[3 marks]

iv.) Compare the speed controlling methods, rotor rheostat control and frequency control. Which method provide the better performance

6. [3 marks]

[Total 20 Marks]

3.

i.) Briefly describe the importance of studying the thermal characteristics of a motor and what are the importance details that can be obtained from temperature rise-time curve

[3 marks]

ii.) Define the **hot-spot temperature** and explain the importance of this parameters when selecting a motor drive

[2 marks]

iii.) Frequency control method is one of the most popular speed controlling methods of a three-phase induction motor and this method is implemented using power electronic converter such as voltage source inverter (VSI). Briefly describe the advantages of using PWM technique to control the VSI to vary the speed of an induction motor

[4 marks]

iv.) Draw the torque-speed curves variation with the frequency control of three phase induction motor. (Draw how the characteristics change with the frequency)

[3 marks]

v.) A 4-pole three phase, induction motor is supplied with 400V,50 Hz, three phase supply. Speed control of the motor is achieved using the frequency control and, in this control scheme, supply voltage to frequency ($^{V}/_{f}$) is maintained at constant value and is equal to the ratio between rated voltage and rated frequency ($^{V_{rated}}/_{f_{rated}}$). Under the rated supply, motor runs at 1440 r.p.m speed.

a.) Find the values of supply frequency and supply voltage required to obtain 600 r.p.m speed [3 marks]

b.) If the supply frequency is changed to 25 Hz, what would be the speed of the motor [3 marks]

c.) What is the range of speeds that can be achieved using this speed control method such that motor can deliver torque up to its maximum possible torque [2 marks]

[Total 20 Marks]

4.

i.) Automatic voltage regulator (AVR) is one of the vital components use with synchronous generator. Briefly describe the requirement of this components and what is the operation of this equipment [5 marks]

ii.) Synchronous motor is not a self-starting motor and we need to provide some starting methods to initiate the motor operations. State two method use to start the synchronous motor and briefly describe them [4 marks]

iii.) State **four** types of toques applied on synchronous motor [4 marks]

iv.) For high power applications, 3-phase synchronous motors are preferred in place of 3-phase induction motor. What is the reason for this choice. [2 marks]

v.) Briefly describe how the reactive power output of the synchronous generator can be controlled and what is the effect of this on power factor of operation [4 marks]

[Total 20 Marks]

5.

i.) Explain the term **duty characteristics** of a motor and briefly explain the importance of this for motor selection for any given application [4 marks]

ii.) Draw the typical temperature rise – time curve for motor operating with short time duty cycle [3 Marks]

iii.) A three-phase induction motor has rated frequency of 50Hz. Can this motor run with 60Hz frequency supply with the same rated power? Give reasons for your answer and explain whether you recommended this selection of motor or not

[3 marks]

iv.) A 6-kW motor has a thermal time constant of 50 minutes and steady state full load temperature rise of 80°C. Full load copper loss to iron loss ratio is 1.25 ($P_{cu,FL}/P_{iron} = 1.25$). Ambient temperature can be considered as 40°C.

a.) What would be the temperature of the motor when it runs for 20 minutes with continuous full load (continuous duty)

[3 marks]

b.) If the motor now uses for short time duty operation and motor run with 7.5 kW load for 10 minutes and rest for 2 hours. What is the temperature rise of the motor when it is operated for 10 minutes under this short-term duty?

[7 marks]

[Total 20 Marks]

6.

i.) Draw a typical current waveform of a synchronous generator under short circuit condition

[4 marks]

ii.) How the voltage unbalance of a three-phase supply affects the performance of a three-phase induction motor. Briefly explain those effects

[4 marks]

iii.) Briefly explain the construction and operation of a bimetallic thermal overload relay

[4 marks]

iv.) Trip class of an overload relay is very important parameter. Briefly describe, how to select an appropriate trip class of an overload relay to provide overload protection for a given three phase induction motor

[4 marks]

v.) Briefly describe the effect of load increase on the operating frequency of a three-phase synchronous generator and how the governor system can use to regulate the frequency at rated frequency of the synchronous machine

[4 marks]

[Total 20 Marks]